CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 15: WRITE THE PYTHON POGRAM TO IMPLEMENT DECISION TREE

AIM

To write a Python program to implement a Decision Tree algorithm for classification using the ID3 algorithm.

ALGORITHM

1. Start the program.
2. Create a dataset containing input attributes and output labels.
3. Calculate the entropy of the dataset.
4. Calculate the information gain for each attribute.
5. Select the attribute with the highest information gain as the decision node.
6. Split the dataset based on the selected attribute.
7. Repeat the process recursively until the data is classified.
8. Display the decision tree and predicted result.
9. Stop the program.

PY8.py - C:/Users/navit/PY8.py (3.12.10)

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```
    for value in set(row[best_attribute] for row in data):
        subset = [
            row for row in data
            if row[best_attribute] == value
        ]

        tree[best_attribute][value] = build_tree(
            subset, remaining, target
        )

    return tree


def predict(tree, sample):
    if not isinstance(tree, dict):
        return tree

    attribute = next(iter(tree))
    value = sample.get(attribute)

    if value not in tree[attribute]:
        return "Unknown"

    return predict(tree[attribute][value], sample)


data = [
    {"Outlook": "Sunny", "Wind": "Weak", "Play": "No"},
    {"Outlook": "Sunny", "Wind": "Strong", "Play": "No"},
    {"Outlook": "Rainy", "Wind": "Weak", "Play": "Yes"},
    {"Outlook": "Rainy", "Wind": "Strong", "Play": "No"},
    {"Outlook": "Overcast", "Wind": "Weak", "Play": "Yes"},
    {"Outlook": "Overcast", "Wind": "Strong", "Play": "Yes"}
]

attributes = ["Outlook", "Wind"]
target = "Play"

tree = build_tree(data, attributes, target)

print("Decision Tree:")
print(tree)

sample = {"Outlook": "Overcast", "Wind": "Weak"}

print("Prediction:", predict(tree, sample))
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/PY8.py ==========
Decision Tree:
{'Outlook': {'Overcast': 'Yes', 'Rainy': {'Wind': {'Strong': 'No', 'Weak': 'Yes'}}, 'Sunny': 'No'}}
Prediction: Yes
>>>
```

RESULT

The Python program was executed successfully, and the Decision Tree was constructed using entropy and information gain to predict the classification.